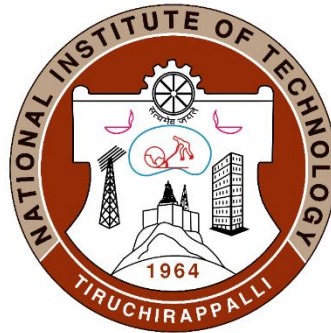


Introduction to Engineering Materials and their properties



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What is Material Science ??

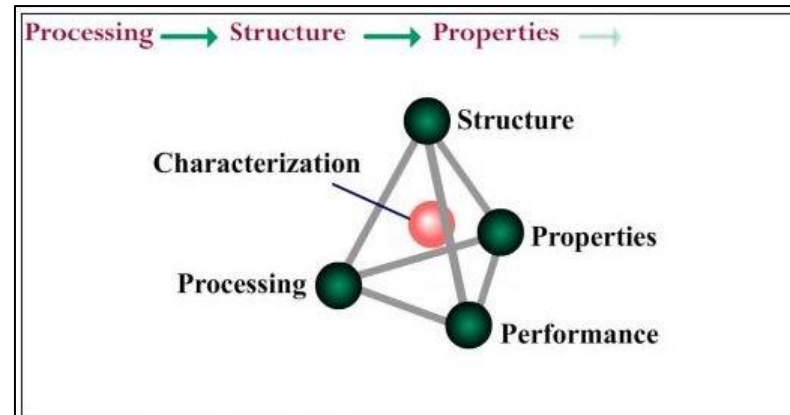


Fig.1 Shows Materials Science Tetrahedron

Correlation between microstructure and properties can be well defined in Material science and engineering

Microstructure depends on the processing route while performance is dictated by properties.

Materials Science and Engineering (MSE)

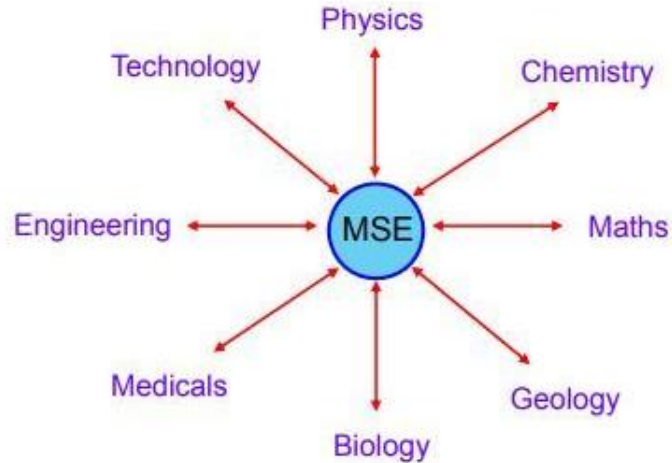


Fig.1 Various areas merge with Materials Science & Engineering

Material Science is an interdisciplinary area where many science and engineering streams merge together

Broad applications of Engineering Materials

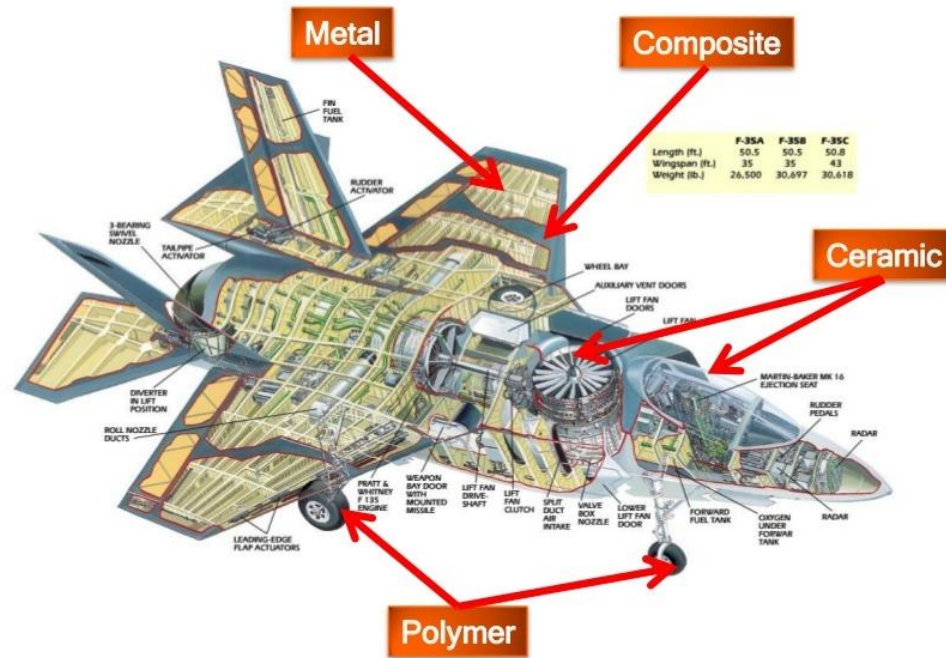


Fig. 3 All type of materials used in Aircraft application

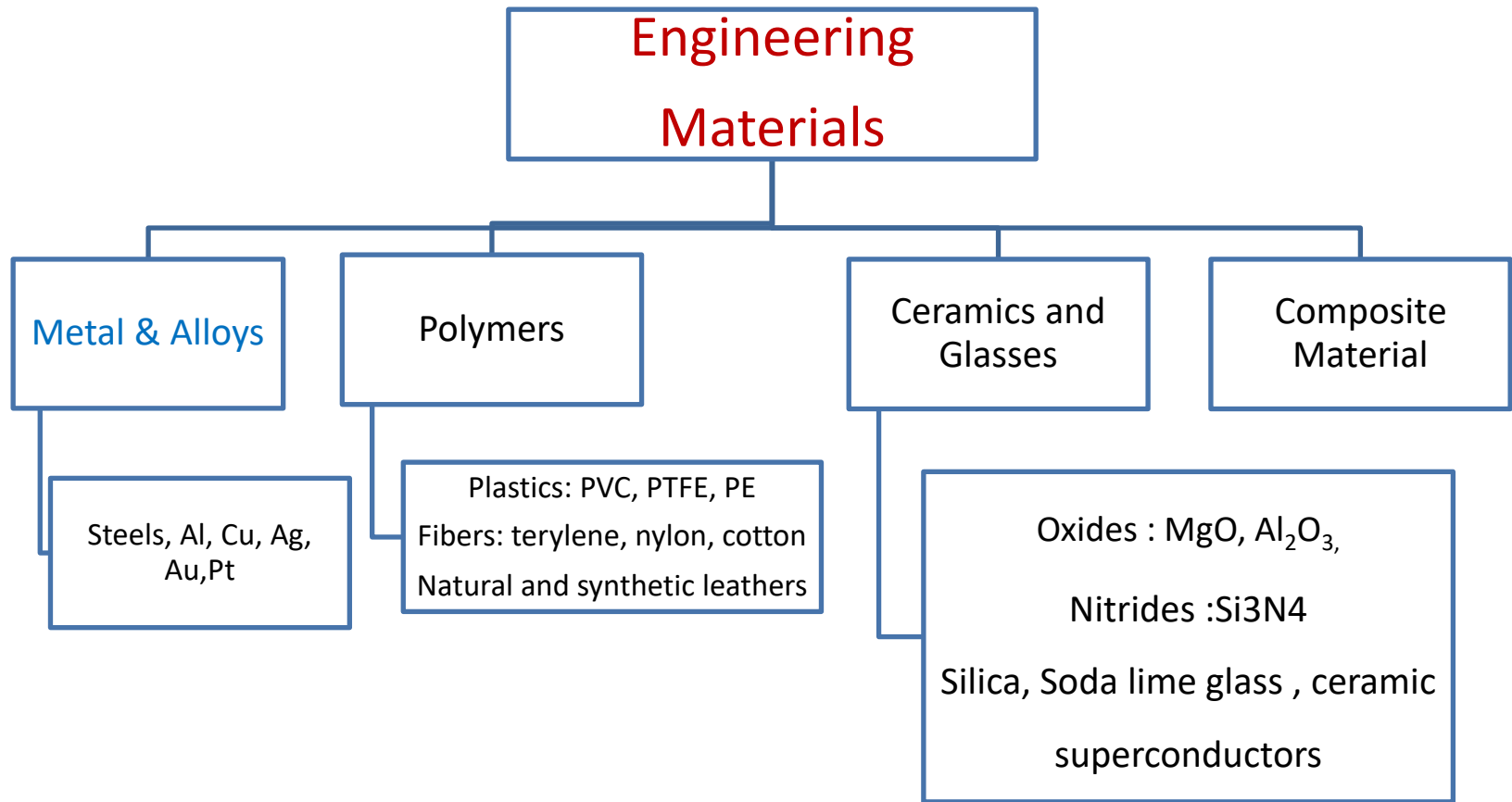


Fig. 4 Broad classification of Engineering Materials

METAL

- A solid material which is typically hard, shiny, malleable, fusible, and ductile, with good electrical and thermal conductivity
- (e.g. iron, gold, silver, and aluminum, and alloys such as steel)



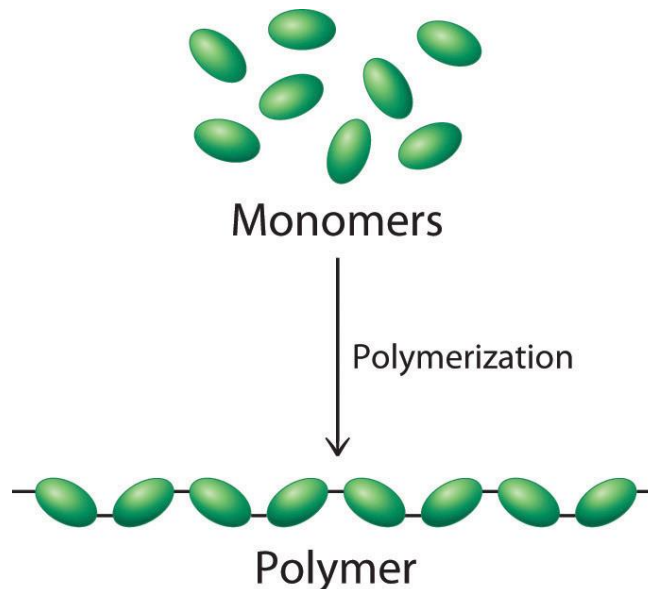
ALLOY

- An **alloy** is a material composed of two or more metals or a metal and a nonmetal.



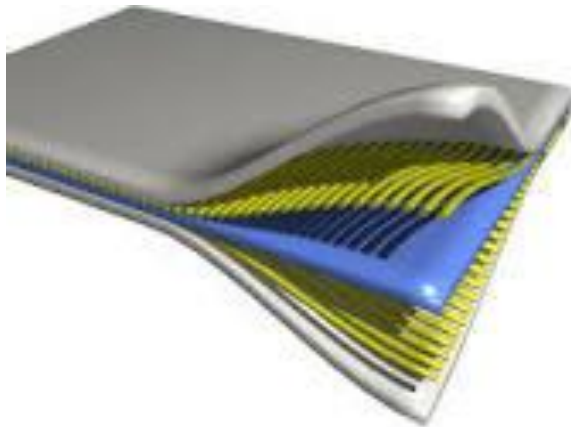
POLYMER

- The word polymer means
- Poly means many and meros means units, parts.
- Polymer means many parts or many units



Composite

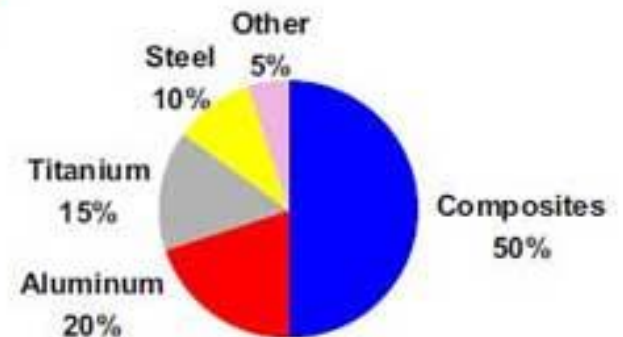
- **Composite** materials (also called composition materials or shortened to **composites**) are materials made from two or more constituent materials with significantly different physical or chemical properties, that when combined, produce a material with characteristics different from the individual components.



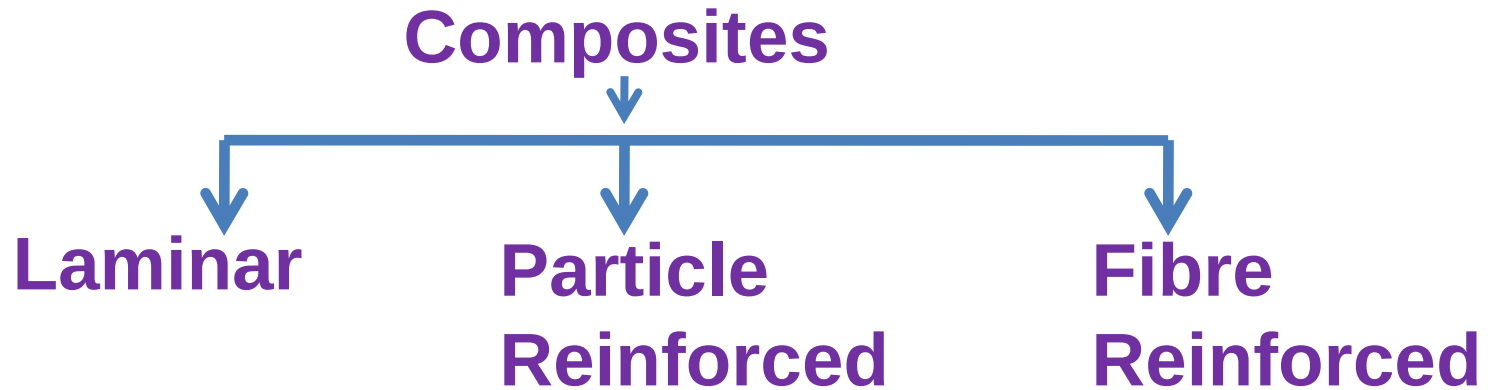
Material used in the Boeing 787



- Carbon laminate
- Carbon sandwich
- Fiberglass
- Aluminum
- Aluminum/steel/titanium pylons



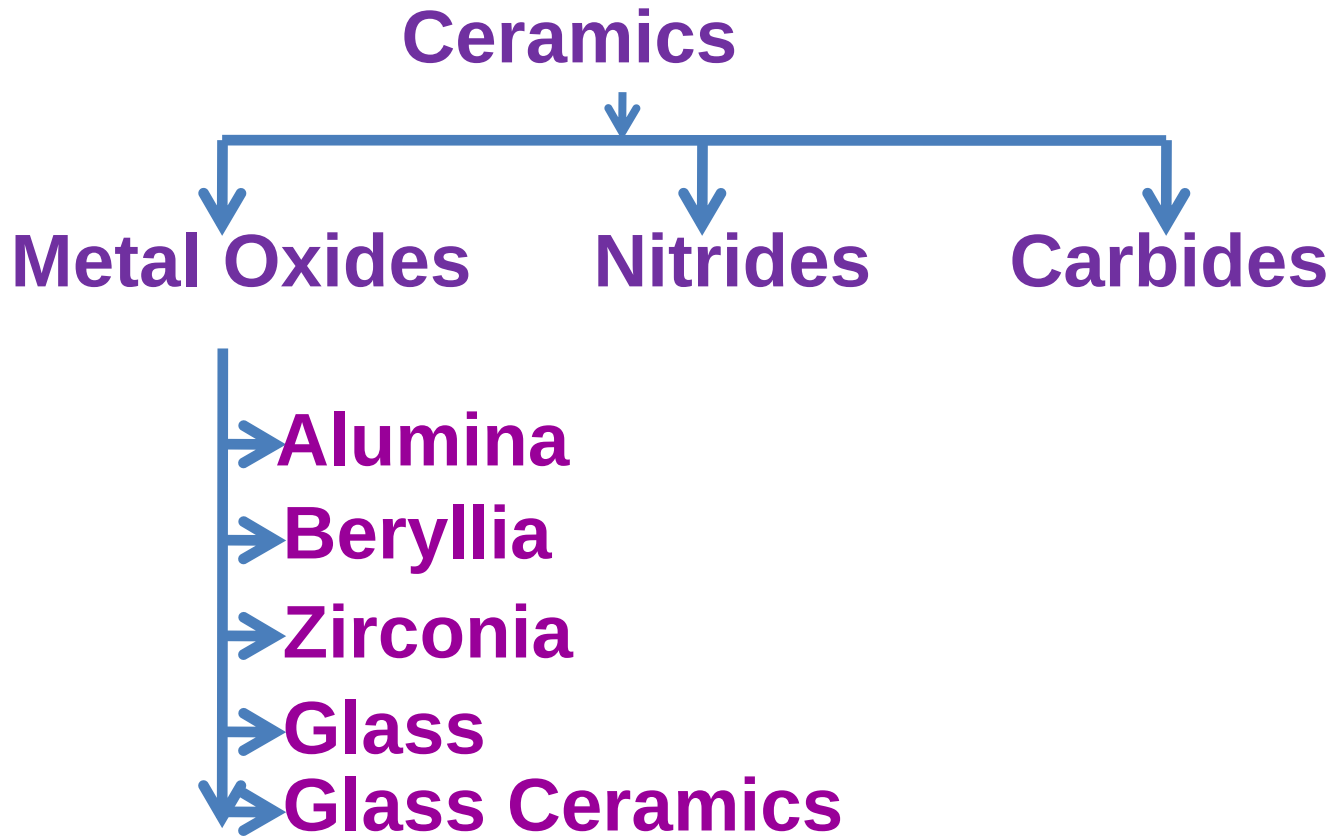
Composites are further classified as:



Ceramics

Ceramics defined as any inorganic (inorganic substances do not contain carbon or its compounds) non metallic material.

Ceramics are further classified as:

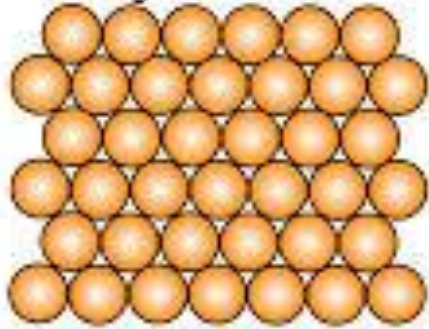




Amorphous material

- The material have no regular arrangement of their molecules.
- Material like glass and paraffin are example of amorphous material.
- These material have properties of solids.
- They have definite volume and shape and diffuse slowly.

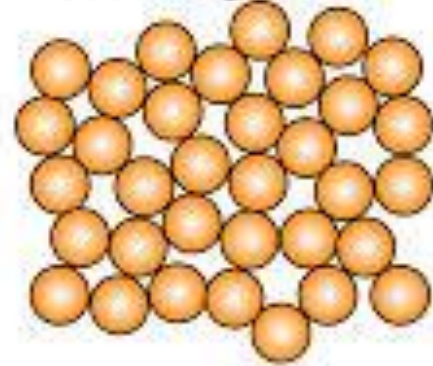
Crystalline



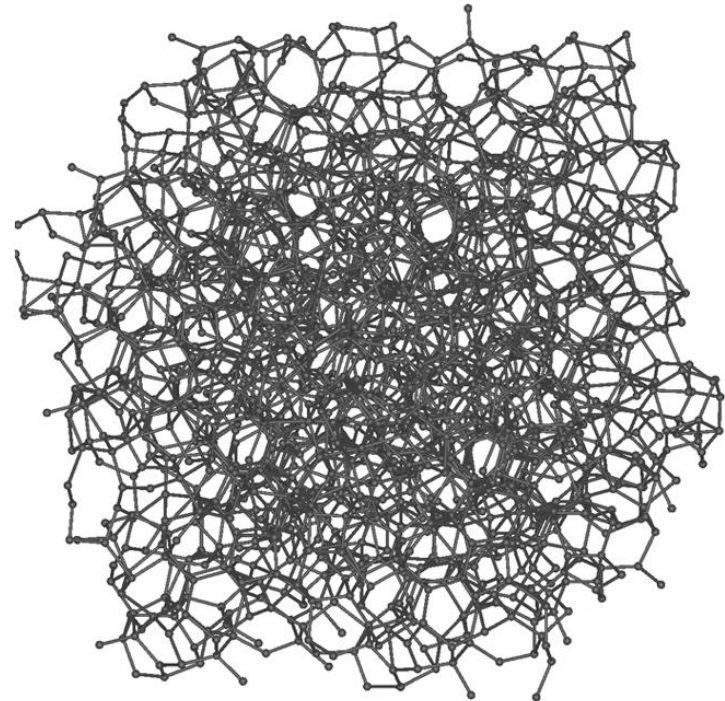
Polycrystalline



Amorphous

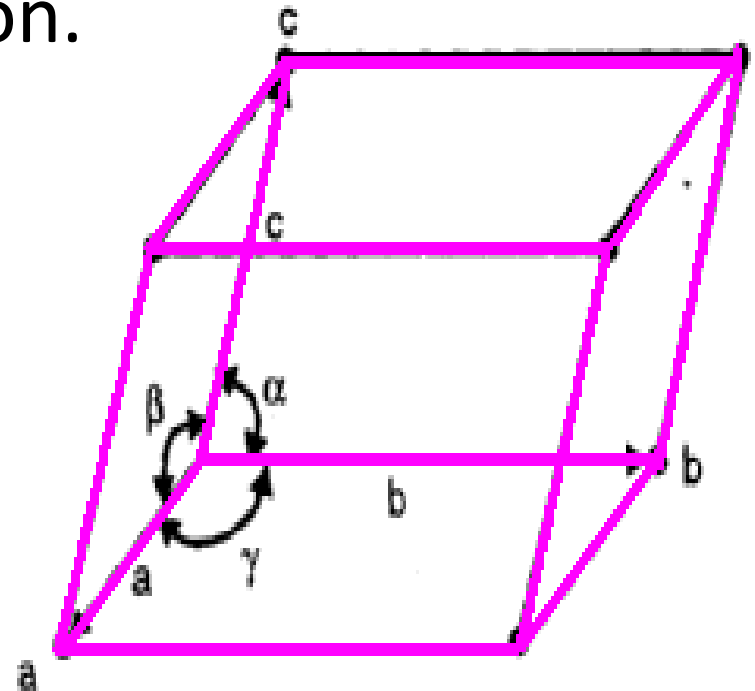
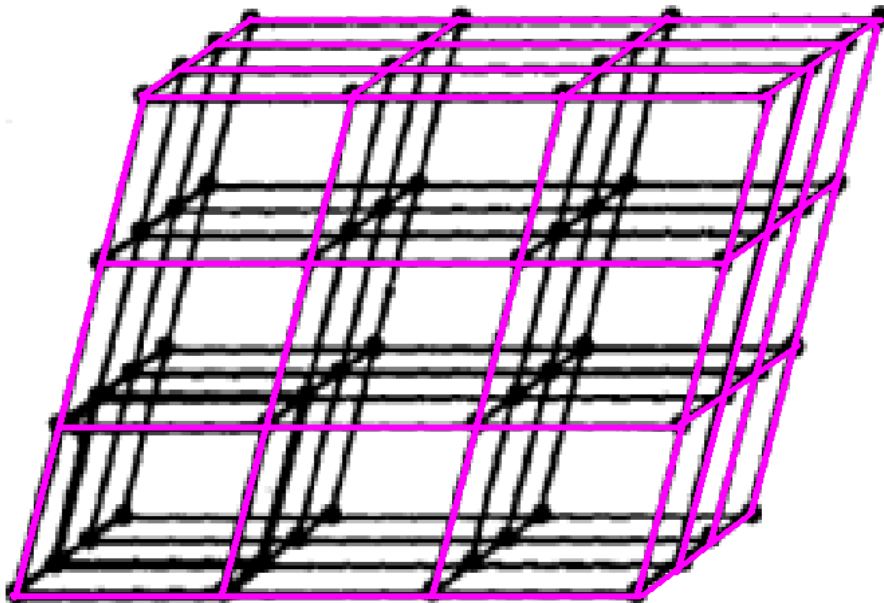


- Amorphous carbon



Crystalline material

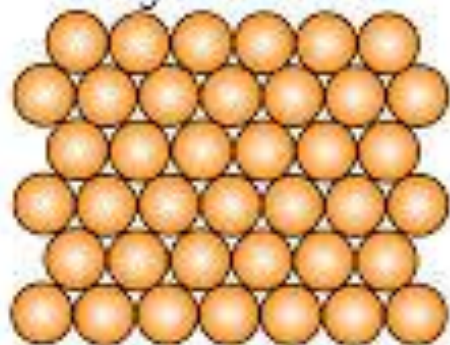
- In a crystalline structure , the atoms are arranged in three dimensional array called a lattice.
- The lattice has a regular repeating configuration in all direction.



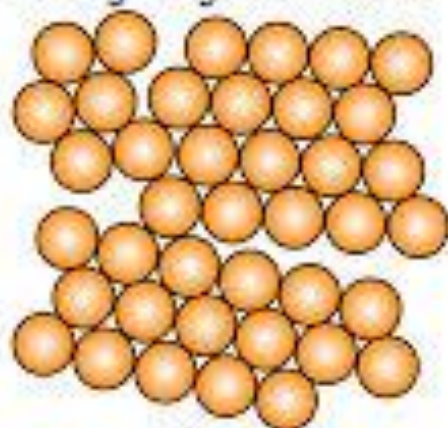
polymorphism

- A crystal is defined as an orderly array of atoms in space.
- Normally metals are made up of number of crystals and each crystal consist of large number of atoms.
- Crystal structure is atomic arrangement in solids.
- Polymorphism is the ability of solid material to exist in more than one form or crystal structure.

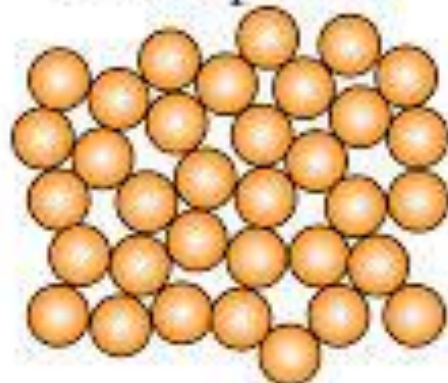
Crystalline



Polycrystalline

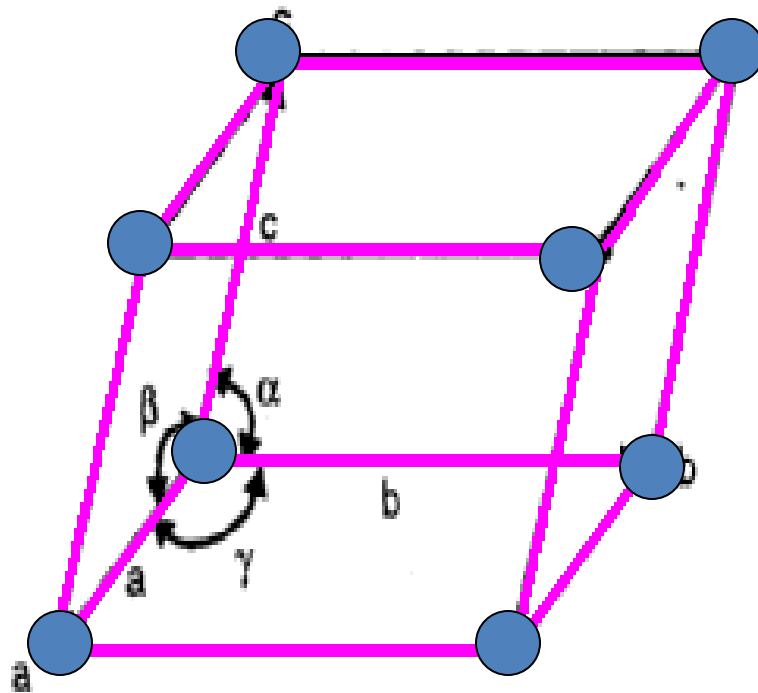


Amorphous

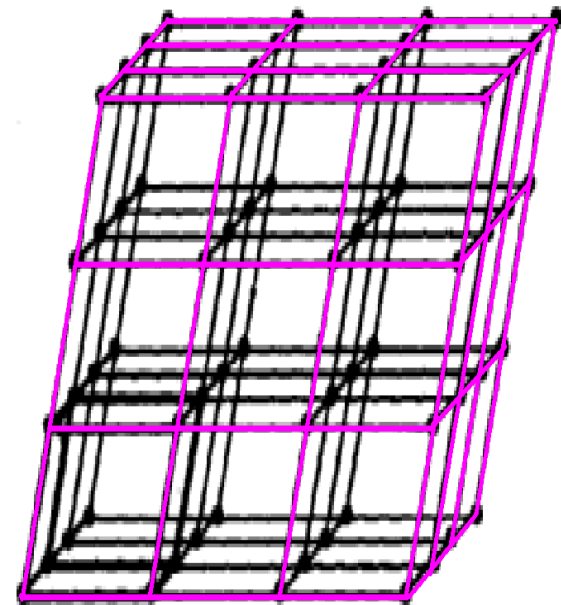




Single atom



Unit Cell



lattice

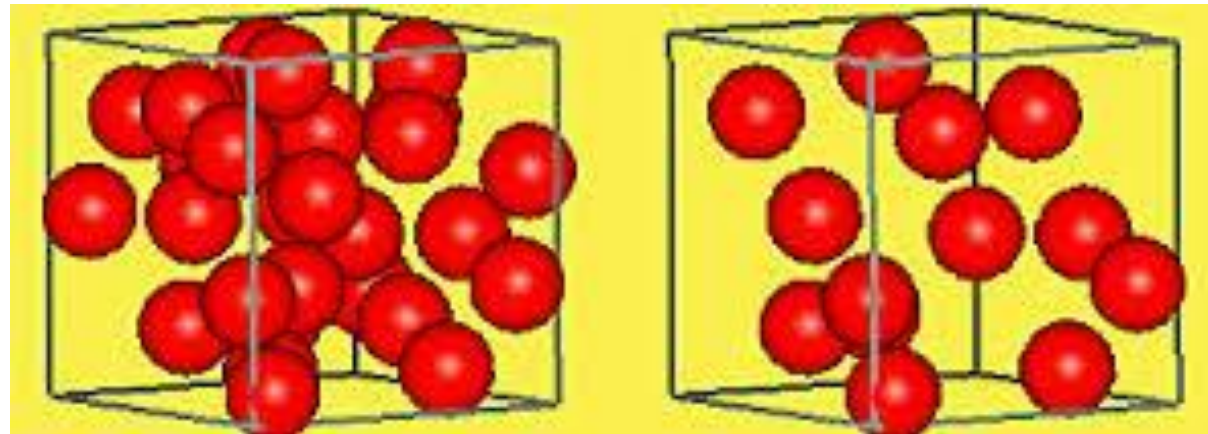
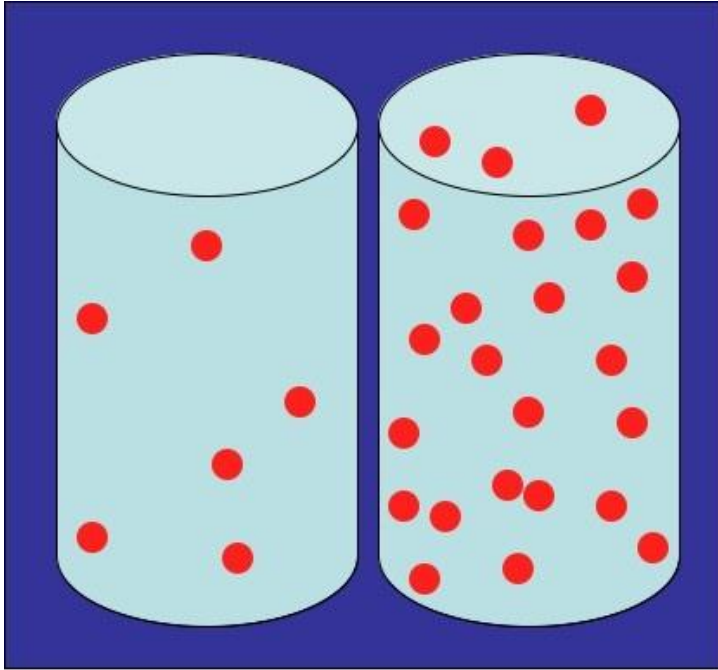
Physical properties

- Density
- Density of a substance is its mass per unit volume.
- The symbol most often used for density is ρ (the lower case Greek letter rho).
- Mathematically, density is defined as mass divided by volume.

$$\rho = \frac{m}{V},$$

- where ρ is the density, m is the mass, and V is the volume.

Density



Melting point

- The **melting point** of a solid is the temperature at which it changes state from solid to liquid at atmospheric pressure.
- The melting point of a substance depends (usually slightly) on pressure and is usually specified at standard pressure.
- The melting point of ice at 1 atmosphere of pressure is very close to 0 °C (32 °F, 273.15 K).



Thermal Properties

ISSUES TO ADDRESS...

- How do materials respond to the application of heat?
- How do we define and measure...
 - heat capacity?
 - thermal expansion?
 - thermal conductivity?
 - thermal shock resistance?
- How do the thermal properties of ceramics, metals, and polymers differ?

Heat Capacity

The ability of a material to absorb heat

- Quantitatively: The energy required to produce a unit rise in temperature for one mole of a material.

heat capacity
(J/mol-K) $\rightarrow$ $C = \frac{dQ}{dT}$

dQ ← energy input (J/mol)

dT ← temperature change (K)

- Two ways to measure heat capacity:

C_p : Heat capacity at constant pressure.

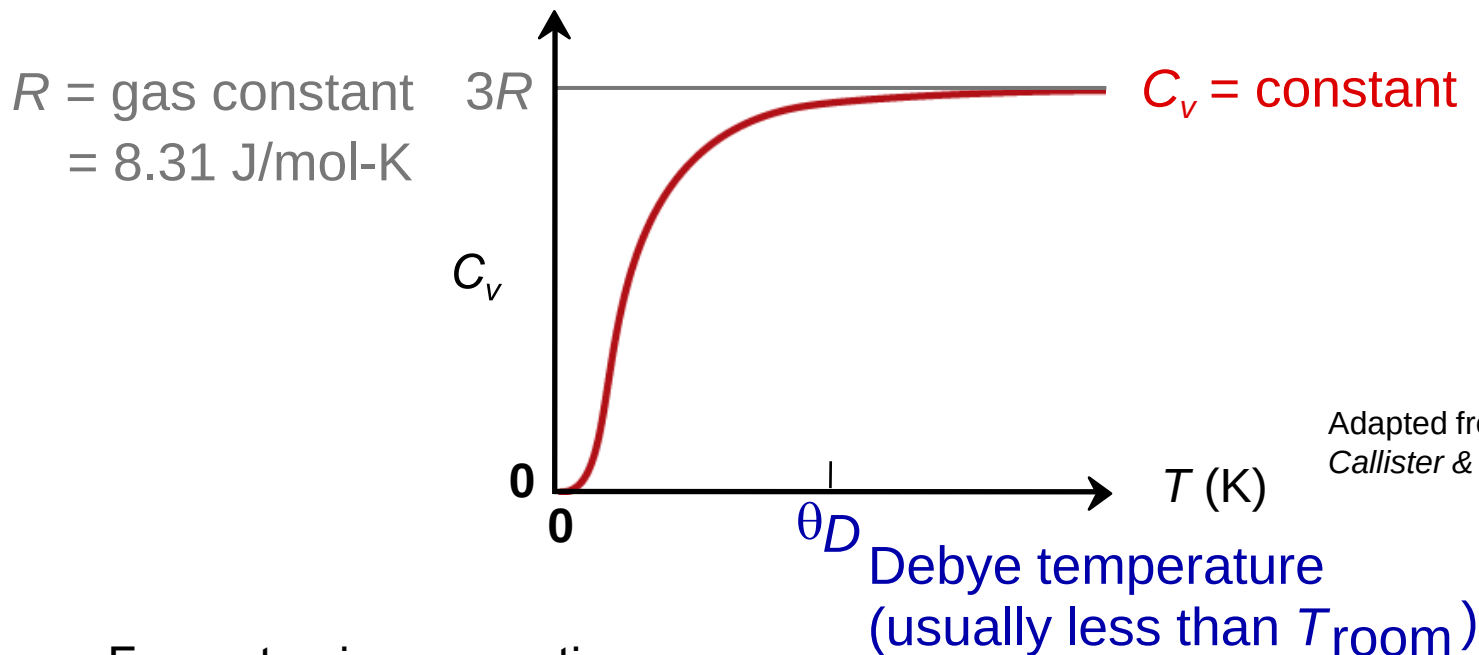
C_v : Heat capacity at constant volume.

$$C_p \text{ usually } > C_v$$

- Heat capacity has units of $\frac{\text{J}}{\text{mol} \cdot \text{K}} \left(\frac{\text{Btu}}{\text{lb-mol} \cdot ^\circ\text{F}} \right)$

Dependence of Heat Capacity on Temperature

- Heat capacity...
 - increases with temperature
 - for solids it reaches a limiting value of $3R$

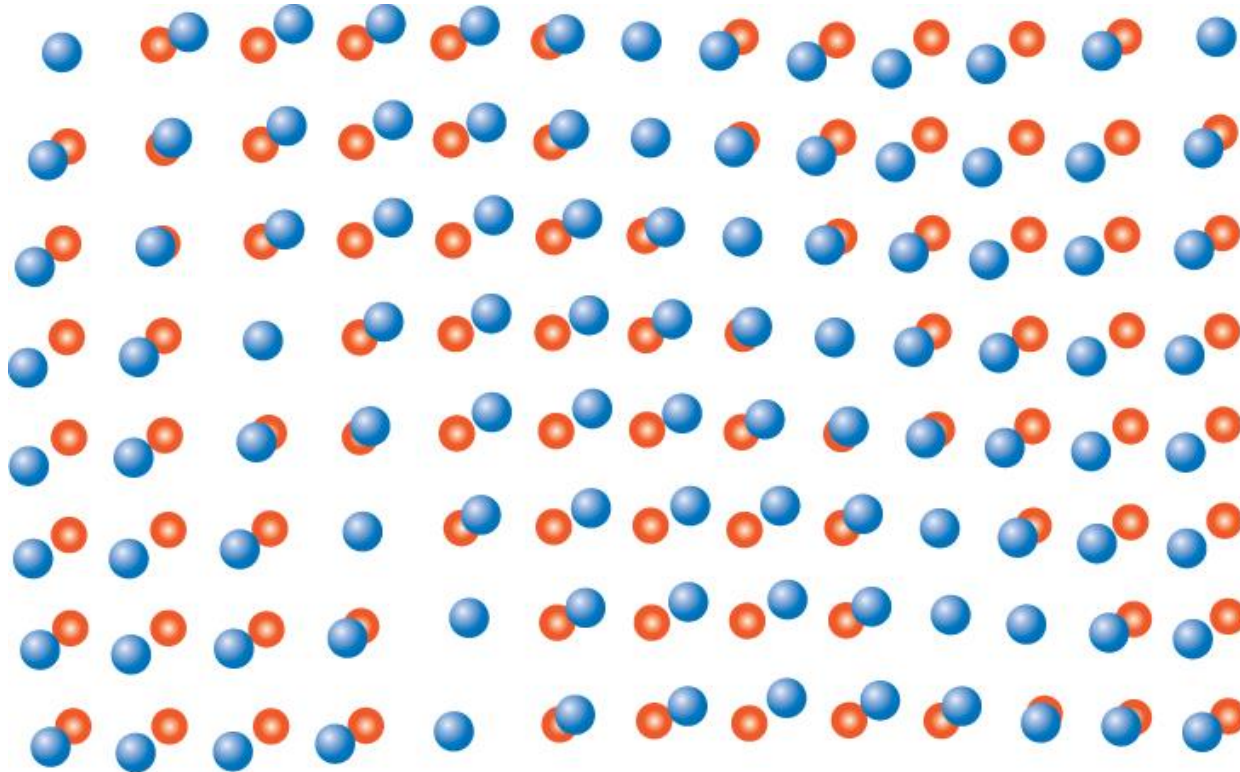


Adapted from Fig. 19.2,
Callister & Rethwisch 8e.

- From atomic perspective:
 - Energy is stored as atomic vibrations.
 - As temperature increases, the average energy of atomic vibrations increases.

Atomic Vibrations

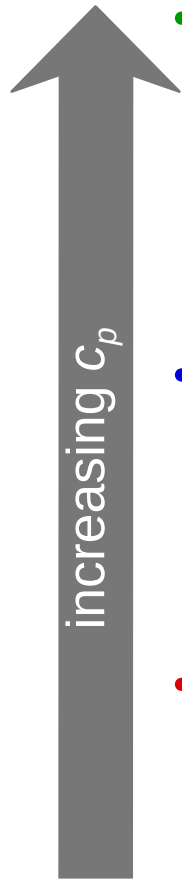
Atomic vibrations are in the form of lattice waves or **phonons**



- Normal lattice positions for atoms
- Positions displaced because of vibrations

Adapted from Fig. 19.1,
Callister & Rethwisch 8e.

Specific Heat: Comparison



Material	c_p (J/kg-K) at room T	
• <u>Polymers</u>		c_p (specific heat): (J/kg-K) C_p (heat capacity): (J/mol-K)
Polypropylene	1925	
Polyethylene	1850	
Polystyrene	1170	
Teflon	1050	
• <u>Ceramics</u>		• Why is c_p significantly larger for polymers?
Magnesia (MgO)	940	
Alumina (Al ₂ O ₃)	775	
Glass	840	
• <u>Metals</u>		
Aluminum	900	
Steel	486	
Tungsten	138	
Gold	128	

Selected values from Table 19.1,
Callister & Rethwisch 8e.

Thermal Expansion

Materials change size when temperature is changed

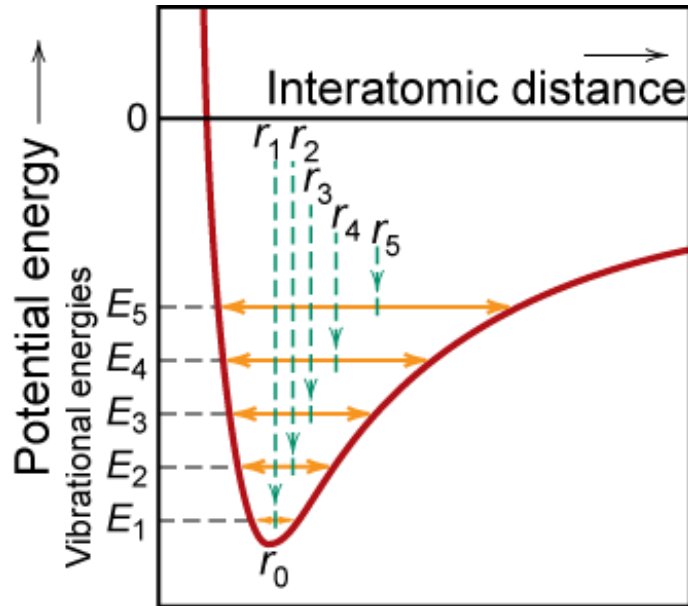


$$T_{\text{final}} > T_{\text{initial}}$$

$$\frac{\ell_{\text{final}} - \ell_{\text{initial}}}{\ell_{\text{initial}}} = \alpha_{\ell} (T_{\text{final}} - T_{\text{initial}})$$

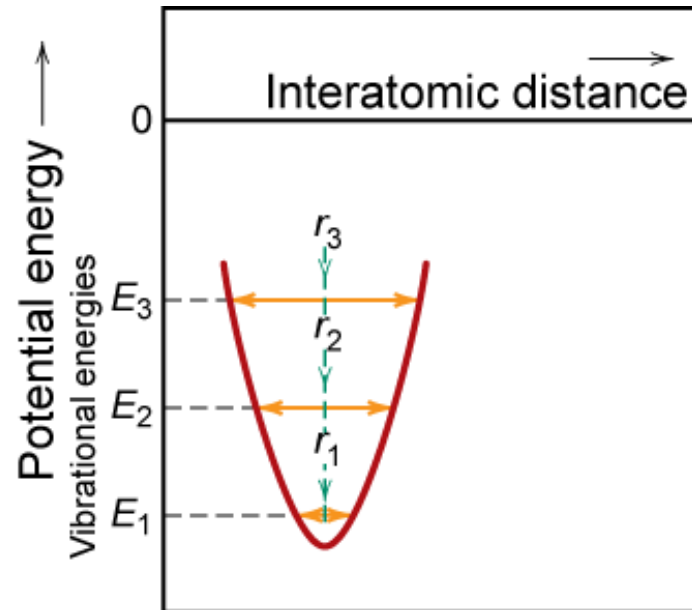
linear coefficient of
thermal expansion (1/K or 1/°C)

Atomic Perspective: Thermal Expansion



Asymmetric curve:

- increase temperature,
- increase in interatomic separation
- thermal expansion



Symmetric curve:

- increase temperature,
- no increase in interatomic separation
- no thermal expansion

Coefficient of Thermal Expansion: Comparison



Material	α_ℓ ($10^{-6}/^\circ\text{C}$) at room T
• <u>Polymers</u>	
Polypropylene	145-180
Polyethylene	106-198
Polystyrene	90-150
Teflon	126-216
• <u>Metals</u>	
Aluminum	23.6
Steel	12
Tungsten	4.5
Gold	14.2
• <u>Ceramics</u>	
Magnesia (MgO)	13.5
Alumina (Al_2O_3)	7.6
Soda-lime glass	9
Silica (cryst. SiO_2)	0.4

Polymers have larger α_ℓ values because of weak secondary bonds

• Q: Why does α_ℓ generally decrease with increasing bond energy?

Selected values from Table 19.1,
Callister & Rethwisch 8e.

Thermal Expansion: Example

Ex: A copper wire 15 m long is cooled from 40 to -9°C . How much change in length will it experience?

- Answer: For Cu

$$\alpha_{\ell} = 16.5 \times 10^{-6} (^{\circ}\text{C})^{-1}$$

rearranging Equation 19.3b

$$\Delta\ell = \alpha_{\ell}\ell_0\Delta T = [16.5 \times 10^{-6} (1/^{\circ}\text{C})](15 \text{ m})[40^{\circ}\text{C} - (-9^{\circ}\text{C})]$$

$$\Delta\ell = 0.012 \text{ m} = 12 \text{ mm}$$

Thermal Conductivity

The ability of a material to transport heat.

Fourier's Law

heat flux (J/m²-s) → $q = -k \frac{dT}{dx}$

temperature gradient

thermal conductivity (J/m-K-s)



- Atomic perspective: Atomic vibrations and free electrons in hotter regions transport energy to cooler regions.

Thermal Conductivity: Comparison



Material	k (W/m-K)	Energy Transfer Mechanism
• <u>Metals</u>		
Aluminum	247	atomic vibrations and motion of free electrons
Steel	52	
Tungsten	178	
Gold	315	
• <u>Ceramics</u>		
Magnesia (MgO)	38	atomic vibrations
Alumina (Al ₂ O ₃)	39	
Soda-lime glass	1.7	
Silica (cryst. SiO ₂)	1.4	
• <u>Polymers</u>		
Polypropylene	0.12	vibration/rotation of chain molecules
Polyethylene	0.46-0.50	
Polystyrene	0.13	
Teflon	0.25	

Selected values from Table 19.1, *Callister & Rethwisch 8e.*

Thermal Stresses

- Occur due to:
 - restrained thermal expansion/contraction
 - temperature gradients that lead to differential dimensional changes

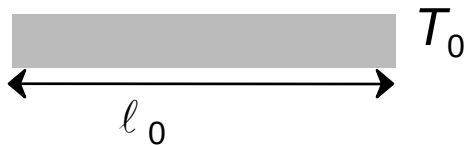
Thermal stress = σ

$$= E\alpha_{\ell}(T_0 - T_f) = E\alpha_{\ell}\Delta T$$

Example Problem

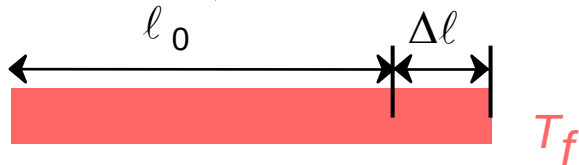
- A brass rod is stress-free at room temperature (20°C).
- It is heated up, but prevented from lengthening.
- At what temperature does the stress reach -172 MPa?

Solution:



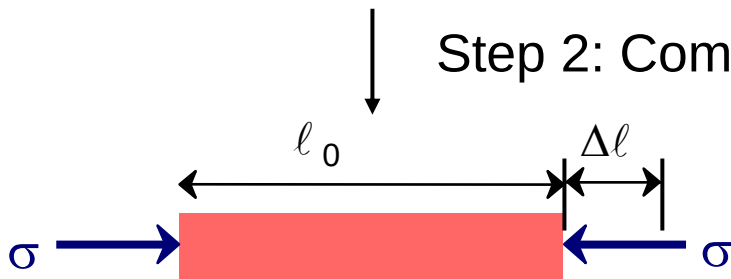
Original conditions

Step 1: Assume unconstrained thermal expansion



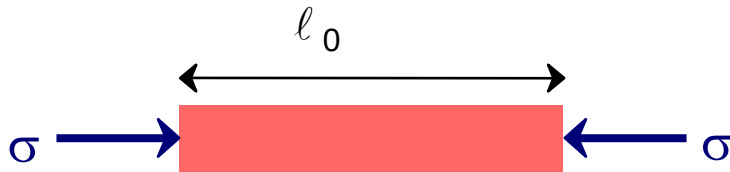
$$\frac{\Delta\ell}{\ell_{\text{room}}} = \epsilon_{\text{thermal}} = \alpha_{\ell}(T_f - T_0)$$

Step 2: Compress specimen back to original length



$$\epsilon_{\text{compress}} = \frac{-\Delta\ell}{\ell_{\text{room}}} = -\epsilon_{\text{thermal}}$$

Example Problem (cont.)



The thermal stress can be directly calculated as

$$\sigma = E(\varepsilon_{\text{compress}})$$

Noting that $\varepsilon_{\text{compress}} = -\varepsilon_{\text{thermal}}$ and substituting gives

$$\sigma = -E(\varepsilon_{\text{thermal}}) = -E\alpha_{\ell}(T_f - T_0) = E\alpha_{\ell}(T_0 - T_f)$$

Rearranging and solving for T_f gives

$$T_f = T_0 - \frac{\sigma}{E\alpha_{\ell}}$$

The diagram illustrates the calculation of the final temperature T_f .

 - T_0 is represented by a pink box with a magenta arrow pointing to it from the label "20°C".

 - σ is represented by a grey box with a grey arrow pointing to it from the label "-172 MPa (since in compression)".

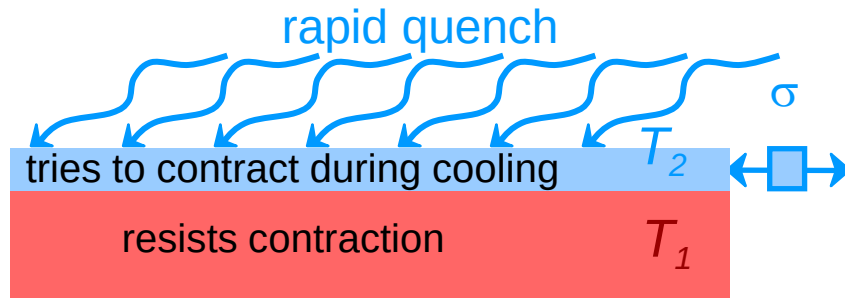
 - E is represented by a green box with a green arrow pointing to it from the label "100 GPa".

 - α_{ℓ} is represented by a blue box with a blue arrow pointing to it from the label "20 x 10⁻⁶/°C".

 The final result is shown in a pink box: **Answer: 106°C**.

Thermal Shock Resistance

- Occurs due to: nonuniform heating/cooling
- Ex: Assume top thin layer is rapidly cooled from T_1 to T_2



Tension develops at surface

$$\sigma = -E\alpha_{\ell}(T_1 - T_2)$$

Temperature difference that can be produced by cooling:

$$(T_1 - T_2) = \frac{\text{quench rate}}{k}$$

Critical temperature difference for fracture (set $\sigma = \sigma_f$)

$$(T_1 - T_2)_{\text{fracture}} = \frac{\sigma_f}{E\alpha_{\ell}}$$

set equal

$$(\text{quench rate})_{\text{for fracture}} = \text{Thermal Shock Resistance (TSR)} \propto \frac{\sigma_f k}{E\alpha_{\ell}}$$

- Large TSR when $\frac{\sigma_f k}{E\alpha_{\ell}}$ is large

Thermal Protection System

- Application:



Chapter-opening photograph, Chapter 23, *Callister 5e* (courtesy of the National Aeronautics and Space Administration.)

- **Silica tiles** (400-1260°C):
 - large scale application

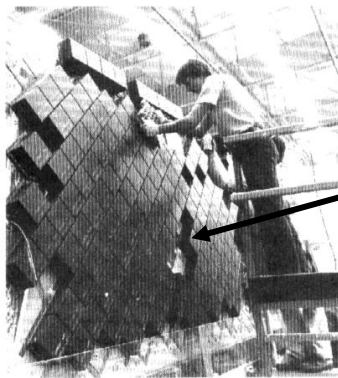


Fig. 19.3W, *Callister 5e*. (Fig. 19.3W courtesy the National Aeronautics and Space Administration.)

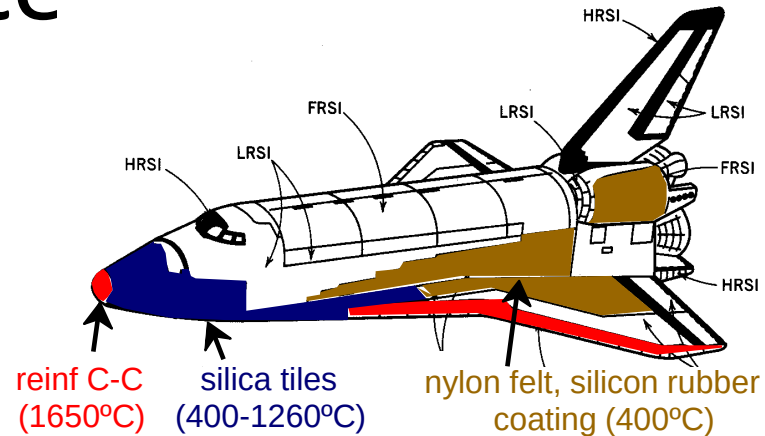
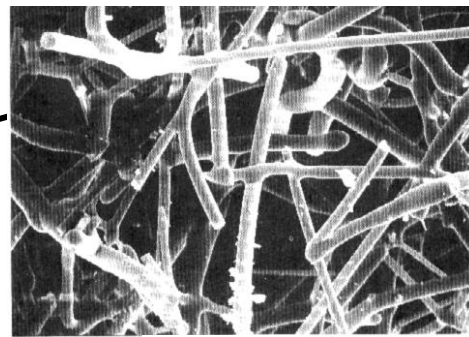


Fig. 19.2W, *Callister 6e*. (Fig. 19.2W adapted from L.J. Korb, C.A. Morant, R.M. Calland, and C.S. Thatcher, "The Shuttle Orbiter Thermal Protection System", *Ceramic Bulletin*, No. 11, Nov. 1981, p. 1189.)

-- microstructure:



~90% porosity!
Si fibers
bonded to one
another during
heat treatment.

← 100 μm →

Fig. 19.4W, *Callister 5e*. (Fig. 219.4W courtesy Lockheed Aerospace Ceramics Systems, Sunnyvale, CA.)

THERMAL DIFFUSIVITY

It measures the ability of a material to conduct thermal energy relative to its ability to store thermal energy. It has the SI unit of m²/s. Thermal diffusivity is usually denoted by α

The formula is:

$$\alpha = k / (c \times \rho)$$

where

k is thermal conductivity (W/(m·K))

ρ is density (kg/m³)

c is specific heat capacity (J/(kg·K))

Summary

The thermal properties of materials include:

- **Heat capacity:**
 - energy required to increase a mole of material by a unit T
 - energy is stored as atomic vibrations
- **Coefficient of thermal expansion:**
 - the size of a material changes with a change in temperature
 - polymers have the largest values
- **Thermal conductivity:**
 - the ability of a material to transport heat
 - metals have the largest values
- **Thermal shock resistance:**
 - the ability of a material to be rapidly cooled and not fracture
 - is proportional to

$$\frac{\sigma_f k}{E \alpha_\ell}$$



Mechanical properties

- Strain
- Change in dimension per unit original dimension is nothing but strain.
- Stress
- Applied force per unit area is nothing but stress.
- $\sigma = F/A$.
- N/mm^2 .

1. Strength

Ability of a material to resist the externally applied force without breaking or yielding.

2. Stiffness

It is ability of material to resist deformation under stress.

3. Elasticity

It is property of material to regain its original shape after deformation when the external force are removed. Steel is more elastic than rubber.

4. Plasticity

It is property of a material which retains the deformation produced under load permanently.

5. Ductility

It is ability of a material enabling it to be drawn in to wire with the application of a tensile force. Steel copper aluminum nickel zinc lead tin are ductile metals

6. Brittleness

It is property of a material opposite to ductility. It is the property of breaking of material with little permanent distortion. Cast iron is a brittle material

7. Malleability

It is special case of ductility which permits materials to be rolled or hammered in to thin sheets. Lead soft steel wrought iron copper aluminum.

8. Toughness

It is property of material to resist fracture due to high impact load like hammer blow.

9. Machinability

It is property of a material which refers to relative ease with which a material can cut.

10. Resilience

It is property of a material to absorb energy and resist shock and impact load.

11. Creep

When a part is subjected to a constant stress at high temperature for long period of time, it will undergo slow and permanent deformation called creep.

12. Fatigue

When a material is subjected to a repeated stresses, it fails at a stresses below the yield point stresses. Such type of failure of material is called fatigue.

13. Hardness

Resistance to wear, scratching, deformation and machinability. It also mean ability of material to cut another metal.

15. Wear : In general, wear is a physical form of material degradation in which the solid material, in small amounts, is removed progressively from the operating surface (s) of bodies that are in relative motion under a mechanical action

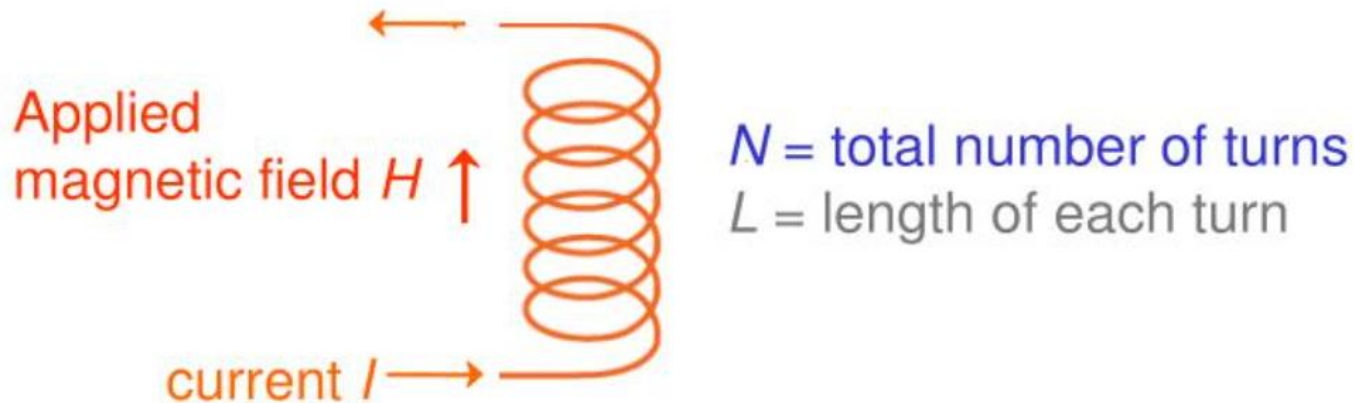
Magnetic properties

Like some metallic materials that are magnetic in nature and display ferromagnetic behaviour, there are some ceramic materials that are magnetic and exhibit ferrimagnetism.

In ferrimagnetic materials, the electron spins are aligned in antiparallel manner but the magnetic moments are unequal.

Applied magnetic field

- Created by current through a coil:



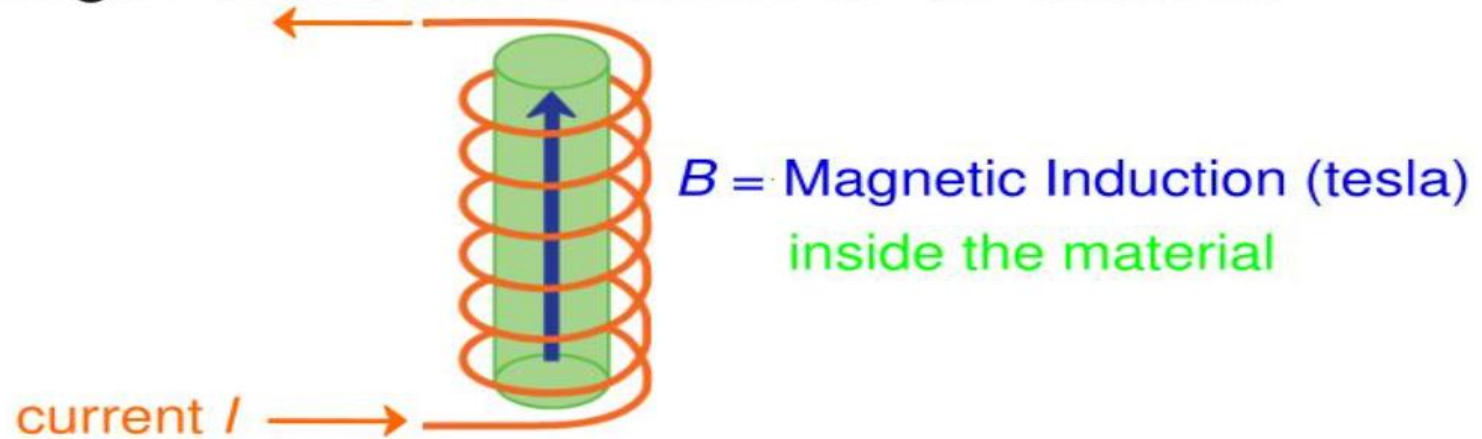
- Relation for the applied magnetic field, H :

$$H = \frac{NI}{L}$$

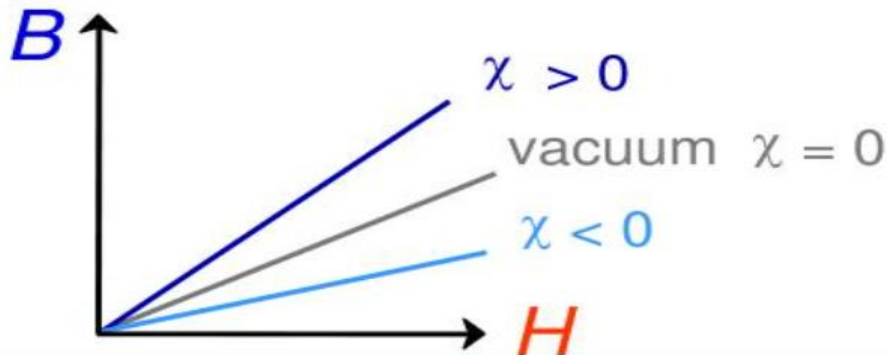
Diagram illustrating the relation for the applied magnetic field H . The equation is shown with the variables H , N , I , and L highlighted in colored boxes (pink, blue, yellow, and grey respectively). An arrow points from the text "current" to the variable I . Below the equation, the text states: "applied magnetic field units = (ampere-turns/m)".

Response to a magnetic field

- Magnetic induction results in the material



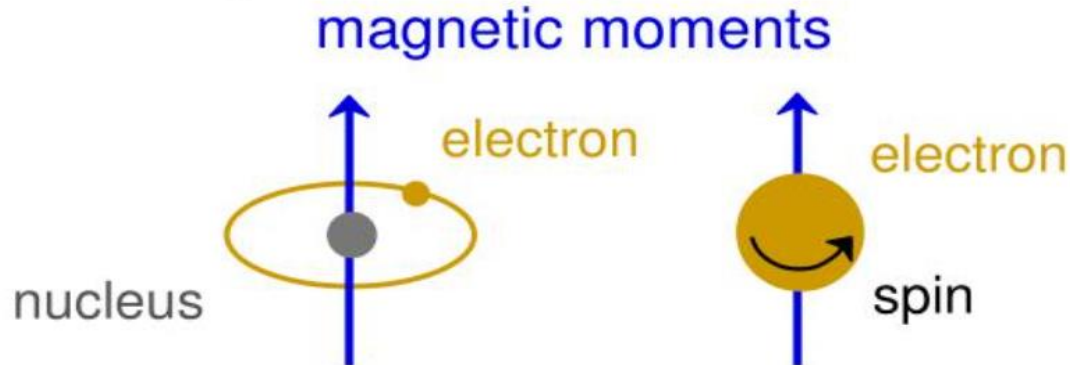
- Magnetic susceptibility, χ (dimensionless)



χ measures the material response relative to a vacuum.

Magnetic Susceptibility

- Measures the response of electrons to a magnetic field.
- Electrons produce magnetic moments:



Adapted from Fig. 20.4,
Callister 7e.

- Net magnetic moment:
--sum of moments from all electrons.
- Three types of response...

Magnetism

- Magnetism is a phenomenon by which a material exerts either attractive or repulsive force on another.
- Basic source of magnetic force is movement of electrically charged particles. Thus magnetic behaviour of a material can be traced to the structure of atoms.
- Electrons in atoms have a planetary motion in that they go around the nucleus. This orbital motion and its own spin cause separate magnetic moments, which contribute to the magnetic behaviour of materials. Thus every material can respond to a magnetic field.
- However, the manner in which a material responds depend much on its atomic structure, and determines whether a material will be strongly or weakly magnetic

Bohr magneton

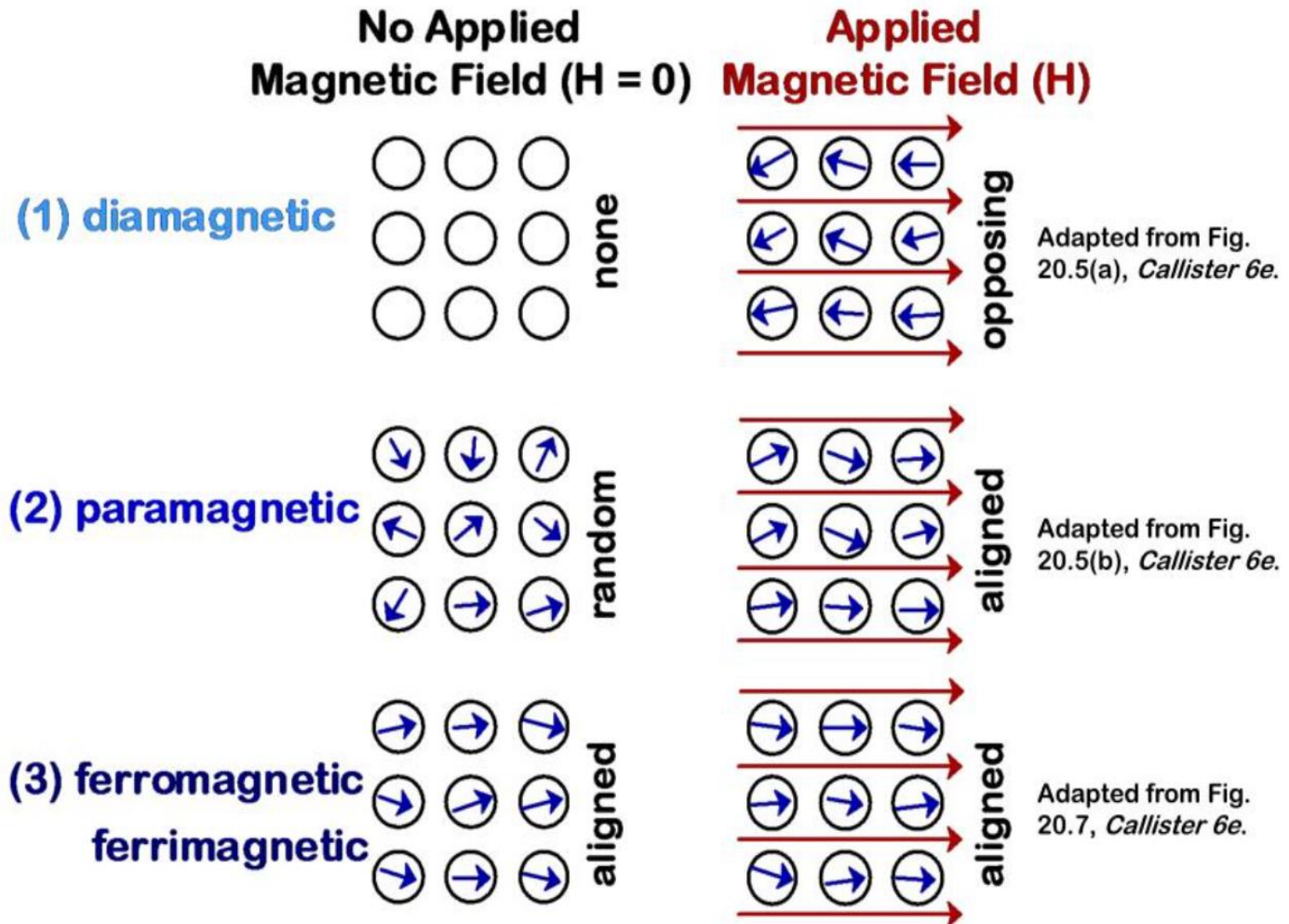
Magnetic moment due to spin of an electron is known as Bohr magneton, M_B .

$$M_B = \frac{qh}{4\pi m_e} = 9.274 \times 10^{-24} \text{ A.m}^2$$

Where q is the charge on the electron, h – Planck's constant, m_e – mass of electron.

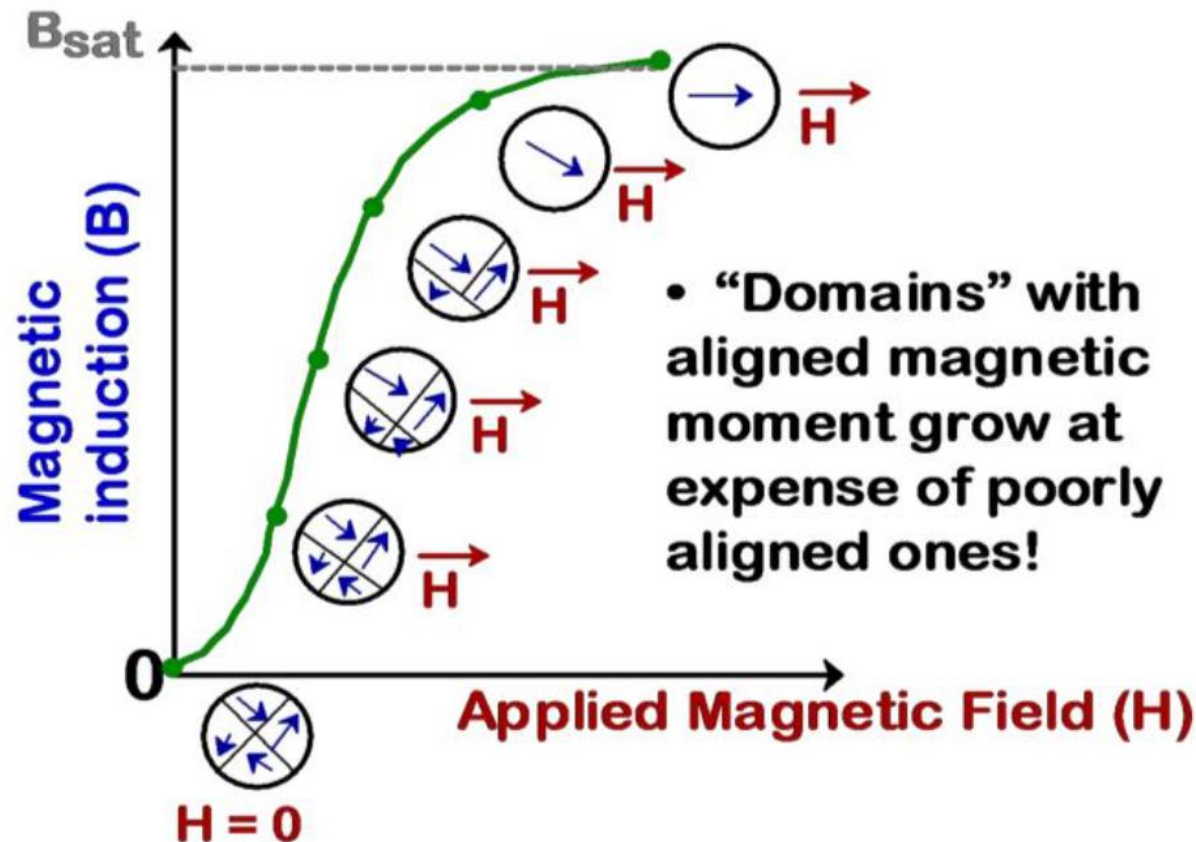
Bohr magneton is the most fundamental magnetic moment.

Magnetic moments for 3 Types



Ferro and Ferri-Magnetic Materials

- As the applied field (H) increases...
--the magnetic moment aligns with H .



Adapted from Fig. 20.13, *Callister 6e*.
(Fig. 20.13 adapted from O.H. Wyatt and D. Dew-Hughes, *Metals, Ceramics, and Polymers*, Cambridge University Press, 1974.)

Permanent Magnets

- **Process:**

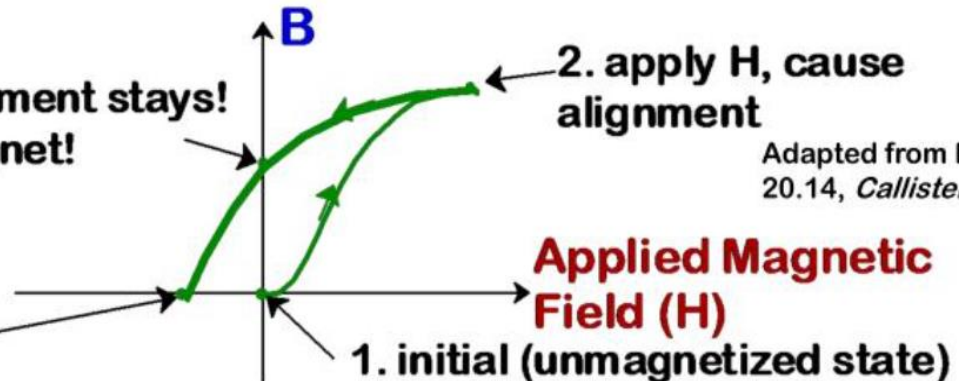
3. remove H, alignment stays!
-> permanent magnet!

2. apply H, cause alignment

Adapted from Fig. 20.14, *Callister 6e*.

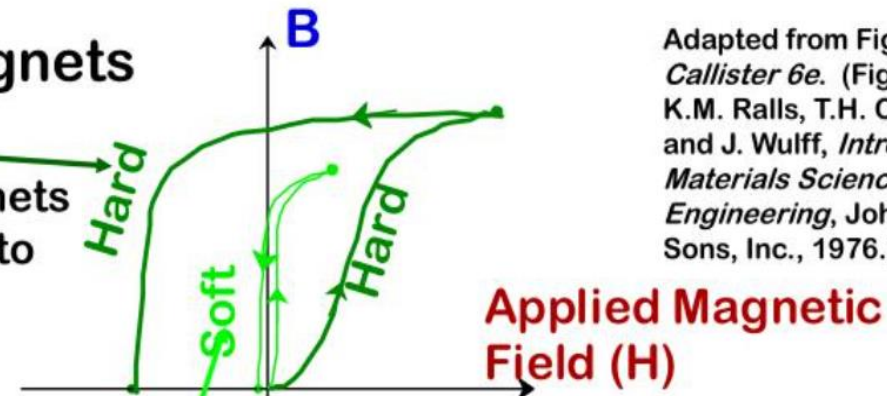
4. Coercivity, H_C :

Negative H needed to demagnetize!



- **Hard vs Soft Magnets**

large coercivity
--good for perm magnets
--add particles/voids to make domain walls hard to move (e.g., tungsten steel:
 $H_C = 5900$ amp-turn/m)



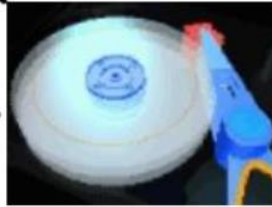
small coercivity--good for elec. motors (e.g., commercial iron 99.95 Fe)

Adapted from Fig. 20.16, *Callister 6e*. (Fig. 20.16 from K.M. Ralls, T.H. Courtney, and J. Wulff, *Introduction to Materials Science and Engineering*, John Wiley and Sons, Inc., 1976.)

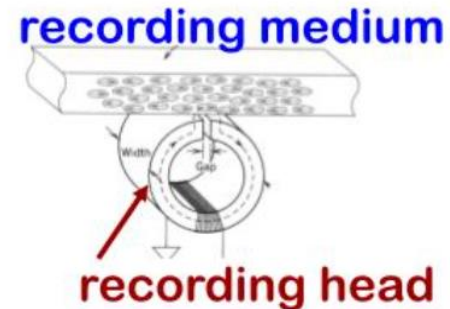
Magnetic Storage

- Information is stored by magnetizing material.
- Head can...

- apply magnetic field H & align domains (i.e., magnetize the medium).
- detect a change in the magnetization of the medium.



Simulation of hard drive
courtesy Martin Chen.
Reprinted with permission
from International Business
Machines Corporation.

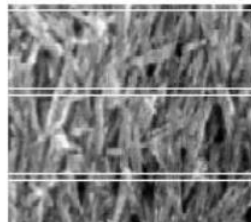


Adapted from Fig. 20.18, *Callister 6e*.
(Fig. 20.18 from J.U. Lemke, *MRS Bulletin*, Vol. XV, No. 3, p. 31, 1990.)

- Two media types:

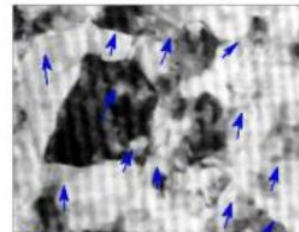
- Particulate: needle-shaped $\gamma\text{-Fe}_2\text{O}_3$. +/- mag. moment along axis. (tape, floppy)

Adapted from Fig. 20.19, *Callister 6e*. (Fig. 20.19 courtesy P. Rayner and N.L. Head, IBM Corporation.)



~2.5 μm ~60nm

- Thin film: CoPtCr or CoCrTa alloy. Domains are ~ 10-30nm! (hard drive)



Adapted from Fig. 20.20(a), *Callister 6e*.
(Fig. 20.20(a) from M.R. Kim, S. Guruswamy, and K.E. Johnson, *J. Appl. Phys.*, Vol. 74 (7), p. 4646, 1993.)

Electrical Properties of Materials

Basic laws and electrical properties of metals

(I)

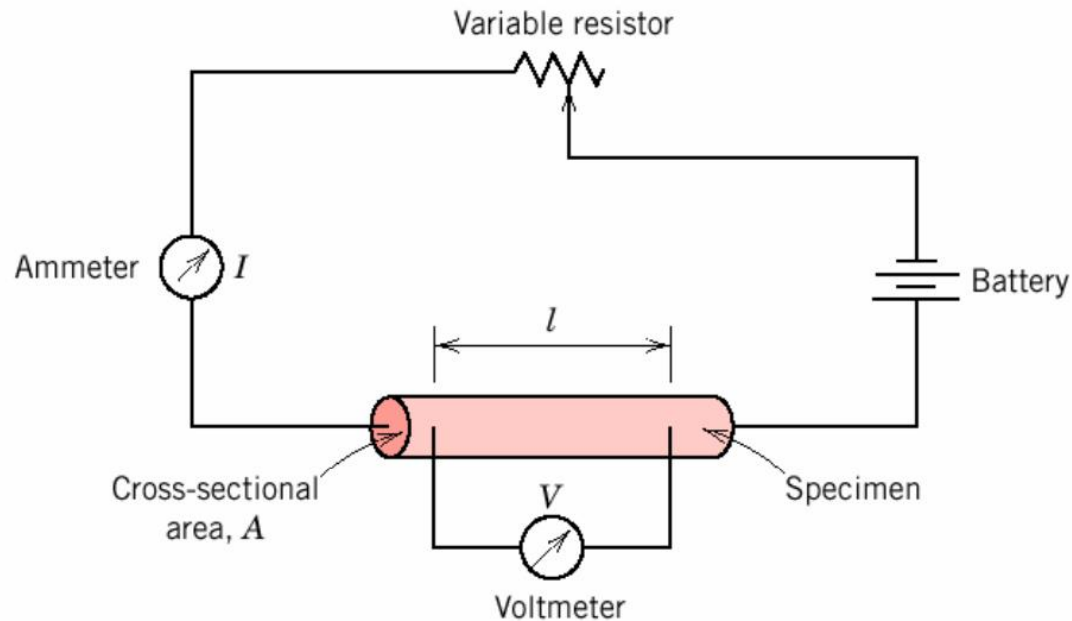
When an **electrical potential** V [volts, J/C] is applied across a piece of material, a **current** of magnitude I [amperes, C/s] flows. In most metals, at low values of V , the current is proportional to V , and can be described by **Ohm's law**:

$$I = V/R$$

where R is the **electrical resistance** [ohms, Ω , V/A].

R depends on the **intrinsic resistivity** ρ of the material [Ωm] and on the geometry (length l and area A through which the current passes):

$$R = \rho l/A$$



In most materials (e.g. metals), the current is carried by electrons (electronic conduction). In ionic crystals, the charge carriers are ions (ionic conduction)

Basic laws and electrical properties of metals (II)

The electrical conductivity (the ability of a substance to an electric current) is the reverse of the resistivity

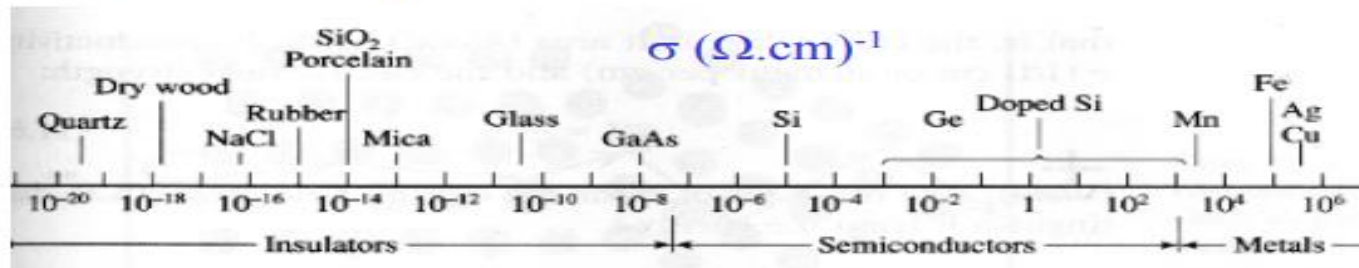
$$\sigma = 1/\rho$$

Since the electric field intensity in the material is

$E = V/l$, Ohm's can be rewritten in terms of the current density $J = I/A$ as

$$J = \sigma E$$

Electrical conductivity varies between different materials by over 27 orders of magnitude, the greatest variation of any physical property



Metals: $\sigma > 10^5 \text{ } (\Omega \cdot \text{m})^{-1}$

Semiconductors: $10^{-6} < \sigma < 10^5 \text{ } (\Omega \cdot \text{m})^{-1}$

Insulators: $\sigma < 10^{-6} \text{ } (\Omega \cdot \text{m})^{-1}$

Conduction in Polymers and Ionic Materials

Ionic Materials

- In ionic materials, the band gap is large and only very few electrons can be promoted to the valence band by thermal fluctuations.
- Cation and anion diffusion can be directed by the electric field and can contribute to the total conductivity: $\sigma_{\text{total}} = \sigma_{\text{electronic}} + \sigma_{\text{ionic}}$
- High temperatures produce more Frenkel and Schottky defects which result in higher ionic conductivity

Polymers

- Polymers are typically good insulators but can be made to conduct by doping.
- A few polymers have very high electrical conductivity - about one quarter that of copper, or about twice that of copper per unit weight



FOR LISTENING

*ANY
QUESTIONS
??*